Model Scope

1. Set and Indices
   - F: set of all flights in the schedule, indexed by f. This is subdivided into F_{arr}(arrivals) and F_{dep}(departures)
   - $S \subseteq F_{arr} \times F_{dep}$: Set of spin turnaround flights (a, d) where a is the arriving flight and d is the departing flight using the exact same physical aircraft on the same stand.
   - V: Set of vehicle classes (e.g., Ambulift, Minibus), indexed by v.
   - M: Set of mutually exclusive operational service modes, m:
     - $m = SA$: Solo Ambulift (Ambulift handles both vertical drop and horizontal transit).
     - $m = CM$: Combined Minibus (Ambulift handles vertical drop; Minibus handles horizontal drive)
     - $m = CP$: Combined Pusher (Ambulift handles vertical drop; Pusher agent walks with passenger)
     - $m = SM$: Solo Minibus (No vertical drop required; Minibus handles horizontal drive).
     - $m = SP$: Solo Pusher (No vertical drop required; Pusher agent walks with passenger).

2. Model Parameters

Financial Parameters

- C_v: Fixed capital expenditure (CapEx) cost of acquiring a vehicle of class v.

Vehicle Capacity Parameters

- $Cap_{v,seat}$: Standard seating capacity of vehicle class v.
- $Cap_{v,wc}$: Dedicated wheelchair tie-down capacity of vehicle class v.

Operational Durations and Buffers

- $\tau_{amb,solo}$: Process duration for a Solo Ambulift run
- $\tau_{amb,comb}$: Process duration for the vertical component of a combined job
- τ_{mini}: Process duration for a minibus transit
- τ_{push}: Process duration for a walking pusher agent assignment
- β: Handover buffer time (the number of minutes a minibus or pusher agent must arrive before the Ambulift completes its vertical drop)

Flight properties and Demand Data

- $D_{f,WCHC}$: Total count of WCHC passengers on flight f.
- $D_{f,WCHS}$: Total count of WCHS passengers on flight f.
- D_{f,own_chair}: Total count of PRM passengers on flight f flagged as has_own_chair. This directly dictates the vehicle wheelchair tie-down capacity requirement ($Cap_{v,wc}$)
- $D_{f,seat}$: Total count of PRM passengers needing standard vehicle seats ($Total\ PRMs - D_{f,own\_chair}$)
- P_f: Total headcount of all PRM passengers on flight f.

- $\omega_{f,m}$: the labour weight (staff count) associated with mode m
- $Staff_{amb}$: The constant number of staff required for to operate an Ambulift
- $Staff_{mini}$: The constant number of staff required for to operate a minbus.
- $Staff_{per\_prm}$: The staff ratio for pusher mode (e.g. 1 staff per PRM)

Categorical Operational Matrix (0 or 1)

- $Remote_f$: Flag indicating the flight is parked on a remote stand
- $EffRemote_f$: Flag indicating the flight requires a vertical lift mechanism
- $Arrival_f$: Flag indicating the flight is an arrival
- $Domestic_f$: Flag indicating the flight belongs to the domestic sector
- $Safety_f$: Flag indicating the flight is on a designated safety stand

SLA Target Parameters

- $T_{chocks,f}$: Actual chocks timestamp for flight f.
- $T_{boarding,f}$: Target timestamp for when departure boarding begins (anchor for departures)
- T_{SLA}: Target SLA window duration (e.g., 20 minutes)
- H_{SLA}: Hard upper bound limit for safe operations (e.g., 60 minutes)

3. Derived and Decision variables

Derived Vertical Lift Requirement Variable

- Before evaluating routing modes, we pre-calculate where flight f physically requires a vertical asset:

$$NeedVertical_f = \begin{cases} 1, & if\ D_{f,WCHC} > 0 \ \vee (EffRemote_f = 1 \ \wedge \ D_{f,WCHS} > 0) \\ 0, & otherwise \end{cases}$$

- The number of sequential vertical transfer batches required to clear flight *f* given the capacity bounds of a single ambulift:

$$Trips_{f,Amb} = \max\left(\left\lceil \frac{D_{f,seat}}{Cap_{Ambulift,seat}} \right\rceil, \left\lceil \frac{D_{f,wc}}{Cap_{Ambulift,wc}} \right\rceil\right)$$

- The number of concurrent minibuses required to satisfy total flight passenger demand:

$$K_{f,Mini} = \max\left(\left\lceil \frac{D_{f,seat}}{Cap_{Minibus,seat}} \right\rceil, \left\lceil \frac{D_{f,wc}}{Cap_{Minibus,wc}} \right\rceil\right)$$

Optimisation Decision Variables

- $N_v \ \in \ \mathbb{Z}^+$: Total number of vehicles of class v maintained in the airport fleet.
- $N_{staff} \ \in \ \mathbb{Z}^+$: Total number of staff required across the terminal to plane to terminal portion of PRM service in a single moment of the day.

- $Staff_f \in \mathbb{Z}^+$: The number of staff members required to service flight *f* based on the chosen mode *m*.
- $x_{f,m} \in \{0,1\}$: Equals 1 if flight f is serviced utilising operational mode $m \in M$; 0 otherwise.
- $t_{start,f}$: Continuous timestamp tracking when servicing begins for flight f.
- $t_{end,f}$: Continuous timestamp tracking when the entire service sequence concludes for flight f.
- $is\_breached_f \in \{0,1\}$: Soft SLA indicator tracking if a service start slips past the target window.
- $h_{a,d} \in \{0,1\}$: Equals 1 if the model elects to hold/retain the arrival Ambulift asset continuously on the stand to cover the departure flight of spin pair (a, d); 0 otherwise.

4. Objective Function

Minimise the global capital investment required to purchase the vehicle fleet capable of satisfying all operational peaks:

$$Min\ \mathrm{Z} = \sum_{v \in V} C_v N_v + (\omega \cdot N_{staff})$$

5. Mathematical Constraints

Service Delivery Framework

Every Flight must be serviced by exactly one operational framework:

$$\sum_{m \in M} x_{f,m} = 1 \quad \forall f \in F$$

Dynamic Flight Duration Linkage

The total duration of a flight's service interval is a direct function of the chosen mode *m*, dynamically scaled by the passenger demand trips:

$$t_{end,f} = t_{start,f} + Duration_f \quad \forall f \in F$$

Where $Duration_f$ is defined as:

$$Duration_f = x_{f,SA} \cdot \left(Trips_{f\,,Amb} \cdot \tau_{amb,solo}\right)$$
$$+ x_{f,CM} \cdot \left(Trips_{f\,,Amb} \cdot \tau_{amb,comb}\right) + \tau_{mini} + \beta$$
$$+ x_{f,CP} \cdot \left(\tau_{amb\,,comb} \cdot \tau_{push}\right)$$
$$+ x_{f,SM} \cdot \tau_{mini}$$
$$+ x_{f,SP} \cdot \tau_{push}$$

Dynamic staffing logic rules

The staff required for flight f $\left(staff_f\right)$ is defined as a function of chose mode *m* and the passenger demand P_f. The total staff requirement is calculated as

$$Staff_f = \sum_{m \in M} x_{f,m} \cdot \omega_{f,m}$$

Where the mode-specific labour coefficient $\omega_{f,m}$ is defined by the following piecewise function:

$$\omega_{f,m} \begin{cases} Staff_{amb} \; if \; m = SA \\ Staff_{mini} \quad if \; m = SM \\ Staff_{mini} + Staff_{amb} \quad if \; m = CM \\ Staff_{amb} + \left(Staff_{per_{prm}} \cdot \; P_f\right) \quad if \; m = CP \\ Staff_{per_{prm}} \cdot \; P_f \quad if \; m = SP \end{cases}$$

Soft SLA Temporal Seach Space

Arrivals (Must start service between chocks and the absolute maximum safety threshold):

$$T_{start,f} \; \geq \; T_{chocks,f} \;\; \forall f \; \in F_{arr}$$

$$T_{start,f} \; \leq \; T_{chocks,f} + \; T_{SLA} + (H_{SLA} - T_{SLA}) \cdot \; is\_breached_f \;\; \forall f \; \in F_{arr}$$

$$T_{start,f} \; \leq \; T_{chocks,f} + \; H_{SLA} \;\; \forall f \; \in F_{arr}$$

Departures (Must complete service before boarding):

$$T_{end,f} \; \leq \; T_{boarding,f} + (H_{SLA} - T_{SLA}) \cdot \; is\_breached_f \;\; \forall f \; \in F_{dep}$$

SLA Breach Logic and Global 98% SLA boundary

$$\sum_{f \in F} is\_breached_f \; \leq \; 0.02 \; \cdot |F|$$

Core physical and Infrastructure Domain Rules

- **Constraint A (Pusher Mode Stand Restriction):** Pusher Agents cannot operate on remote stands where walking to the terminal is impossible:

$$\sum_{m \in \{CP,SP\}} x_{f,m} \; \leq 1 - \; Remote_f \quad \forall \, f \; \in F$$

- **Constraint B (Passenger-Driver Vertical Requirement):** The chosen operational routing (m) must strictly align with the passenger-driver vertical requirement metric:

$$\sum_{m \in \{SA,CM,CP\}} x_{f,m} = \; NeedVertical_f \quad \forall \, f \; \in F$$

$$\sum_{m \in \{SM,SP\}} x_{f,m} = \; 1 - \; NeedVertical_f \quad \forall \, f \; \in F$$

- **Constraint C (Domestic Arrival Ambulift Exclusion):** Ambulifts are logistically banned from performing horizontal and arrival movements for domestic sectors due to customs/terminal design constraints:
$$x_{f,SA} \leq 1 - \left(Arrival_f \cdot Domestic_f\right) \;\; \forall f \in F$$
- **Constraint D (Safety Stand Pusher Restrictions):** Combined Pusher agents are prohibited from operating on safety stands. Solo pushers are permitted.
$$\sum_{m \in \{CP\}} x_{f,m} \leq 1 - Safety_f \;\; \forall f \in F$$

Rostering/Interval Capacity Tracking constraints

Let I_f represent the active scheduling interval variable for flight *f*, spanning from $\left[t_{start,f},\;\; Duration_f, t_{end,f}\right]$.

To compute the fleet capacities (N_{Amb}, N_{mini}) and peak staffing (N_{staff}) dynamically without explicit timeline looping indices, the resources consumed across overlapping intervals at any given moment time (*t*) are constrained as follows:

Ambulift Capacity Restriction

Only **one** Ambulift can be attached to an aircraft at a time. If a vertical mode is selected, it consumes 1 resource slot. Additionally, if an Ambulift is held continuously for a spin turnaround pair ($h_{a,d} = 1$), it remains unavailable for other flights during the intermediate turnaround:

$$\sum_{f \in F} \left(x_{f,SA} + x_{f,CM} + x_{f,CP}\right) \cdot \mathbb{I}\left(t \in I_f\right) + \sum_{(a,d) \in S} h_{a,d} \cdot \mathbb{I}\left(t_{end,a} \leq t < t_{start,d}\right) \leq N_{ambulift} \;\; \forall t$$

Where $\mathbb{I}$ is an indicator function evaluating to 1 if the condition is met at time *t,* and *0* otherwise).

Minibus Capacity Restriction

Minibuses scale concurrently to match passenger volumes when a minibus-supported mode (*CM* or *SM*) is chosen:

$$\sum_{f \in F} \left(x_{f,CM} + x_{f,SM}\right) \cdot K_{f,Mini} \cdot \mathbb{I}\left(t \in I_f\right) \leq N_{Minibus} \quad \forall t$$

Global Staff Capacity Restriction

The global staffing ceiling tracks the concurrent sum of all active labour required across all flight jobs at peak times:

$$\sum_{f \in F} Staff_f \cdot \mathbb{I}\left(t \in I_f\right) \leq N_{staff} \quad \forall t$$

Spin Turnaround Scheduling Metrics

- **Turnaround Sequence Rule:** Passengers cannot board a departing spin flight until the corresponding arriving flight is entirely clear:

$$t_{start,d} \geq t_{end,a} \;\; \forall (a,d) \in S$$